



Systematic Review

Designing Retail Central Bank Digital Currencies: A Systematic Literature Review of Trade-Offs Between Security, Privacy, and Financial Stability

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Abstract

This paper proposes a CBDC design trilemma, the claim that central banks cannot simultaneously maximize privacy, financial stability, and regulatory compliance when designing retail central bank digital currencies and finds the existing literature consistent with this proposition. Through a systematic review of 140 peer-reviewed articles (Web of Science SCIE/SSCI indexes, 2014–2026, supplemented by Scopus and SSRN), evidence is synthesized across four thematic dimensions: design frameworks and architecture, financial stability and banking risk, privacy and security trade-offs, and user adoption and institutional quality. Cross-tabulation of coded data supports all three pairwise tensions: privacy-enhancing designs weaken AML/CFT enforcement, anonymous holdings amplify bank-run risk, and stringent prudential safeguards constrain transaction monitoring. The literature converges on two-tier, hybrid architectures with tiered privacy as the dominant compromise a “zone of feasible design”, that sacrifices full optimality on each vertex. Nine research gaps are identified, most critically the scarcity of empirical evidence from live deployments, the neglect of wholesale CBDC, and insufficient analysis of cross-border interoperability. The framework offers policymakers a structured lens for evaluating retail CBDC design trade-offs and researchers a testable proposition for future empirical work.

Keywords: CBDC; central bank digital currency; privacy; financial stability; user adoption; digital currency design; financial inclusion; trade-offs; security



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1. Introduction

The global shift toward digital money is reshaping the foundations of financial systems. Cash usage in everyday transactions has steadily declined, yet demand for cash as a store of value has remained surprisingly resilient, a phenomenon described as the “cash paradox” (Jiang & Shao, 2019). Simultaneously, private digital platforms and cryptocurrencies have challenged the central role of state money (Bai et al., 2025). As Davoodalhosseini and Rivadeneyra (2020) caution, “leaving the provision of safe and efficient means of payment to the market may not result in an efficient and stable outcome” (Davoodalhosseini & Rivadeneyra, 2020). Against this backdrop, central banks worldwide are exploring the issuance of central bank digital currencies (CBDCs). As of mid 2024, over 130 countries representing 98% of global GDP were investigating CBDC projects at various stages of

research, development, and deployment ([Atlantic Council, 2025](#)). Landmark pilot programmes including China's e-CNY, the Bahamas' Sand Dollar, Nigeria's eNaira, and the European Central Bank's Digital Euro investigation have moved CBDCs from theoretical concept to practical policy consideration.

However, CBDC design is not a straightforward technical exercise. Each design choice creates trade-offs across multiple dimensions: an architecture that maximizes privacy may undermine regulatory compliance; features that protect financial stability may reduce adoption appeal; and mechanisms that promote financial inclusion may introduce new security vulnerabilities. This paper argues that these tensions appear to be structural within the current literature, that they constitute a CBDC design trilemma whose edges and implications are developed in Section 2 and evaluated against the evidence in Sections 4 and 5.

Despite a rapidly growing body of literature, existing reviews tend to focus on single dimensions such as financial stability implications ([Keister & Sanches, 2022](#)), technical architecture ([Auer & Böhme, 2020](#)), or adoption factors ([Jabbar et al., 2023](#)) without systematically mapping the cross-dimensional trade-offs that define the CBDC design challenge. No prior review has synthesized the evidence base to identify the structural constraints that bind these dimensions together. This review fills that gap by proposing a unifying framework for the CBDC design trilemma and systematically evaluating it against 140 coded articles.

1.1. Research Questions

This review is guided by one primary research question (PRQ) and three sub-questions:

PRQ: How do central bank digital currency (CBDC) design choices influence the trade-offs between user adoption, privacy, financial inclusion, and financial stability?

1. SQ1: What factors most strongly drive or hinder public adoption of CBDC across different jurisdictions?
2. SQ2: How do privacy–security trade-offs affect trust and willingness to adopt CBDC?
3. SQ3: Which CBDC design architectures best support financial inclusion without destabilizing commercial banking?

1.2. Contribution

This study makes three contributions. First, it proposes a CBDC design trilemma, the claim that no retail CBDC architecture can simultaneously maximize privacy, financial stability, and regulatory compliance, and finds the coded evidence of 140 peer-reviewed articles to be consistent with each pairwise tension. Second, it provides the most comprehensive cross-dimensional systematic review of CBDC literature to date, with quantitative synthesis (cross-tabulations, vote counts, odds ratios) that moves beyond narrative description. Third, it demonstrates how the trilemma operates differently across jurisdictional contexts, offering policymakers a structured framework for navigating design choices within their specific constraint space.

1.3. Prior Systematic Reviews of the CBDC Literature

The CBDC literature has attracted several prior review efforts, yet none provide a comprehensive, PRISMA-compliant synthesis across all four design dimensions examined in this study. Understanding where existing reviews fall short is necessary to establish the contribution of the present work.

[Hoang et al. \(2023\)](#) conducted the first systematic review of the CBDC literature using a text-mining approach, analysing 191 abstracts from Scopus up to February 2022. Using topic modelling and frequency analysis, they identified seven broad research clusters,

including CBDC design and technology, monetary policy, payment systems, and financial stability. While valuable for mapping the topography of the field, this study has important limitations: the corpus is restricted to Scopus and predates the substantial expansion of the literature after 2022. More fundamentally, the text-mining method extracts co-occurrence patterns in abstracts rather than evaluating study findings; the authors do not assess the quality, methodology, or conclusions of individual studies, nor do they identify cross-dimensional trade-offs.

Dionysopoulos et al. (2023) offer a critical and interdisciplinary review covering motivations for CBDC issuance, design options, and monetary policy implications. Unlike earlier surveys, their review explicitly evaluates the arguments in the literature rather than merely cataloguing them, spanning finance, economics, and computer science. However, the review is not PRISMA-compliant and does not follow a structured protocol for study selection or data extraction. As the authors acknowledge, it prioritizes the “critical element” over systematic coverage and does not include user adoption and institutional trust as a distinct analytical dimension, nor does it propose a framework connecting design choices to outcomes across dimensions.

Chen and Siklos (2022) review macro-financial challenges surrounding CBDC, focusing specifically on the implications of a narrow CBDC replacing notes and coins and a broader deposit-bearing form for inflation and financial stability. Their review is explicitly scoped to macroeconomic implications and supplemented by empirical simulations using historical monetary data. While rigorous within its scope, it does not address security architecture, privacy trade-offs, or adoption determinants, and does not constitute a systematic review under PRISMA criteria.

Elsayed and Nasir (2022) provide a forward-looking research agenda for CBDC studies, identifying understudied areas and proposing directions for future empirical and theoretical work. This is agenda-setting rather than synthetic: it maps what remains unknown rather than consolidating what is known. Similarly, Auer et al. (2022) survey CBDC motivations and the research frontier from an economics perspective, covering cross-border payments, financial inclusion, and monetary policy but without a systematic protocol or coverage of the adoption or privacy dimensions.

Taken together, the existing reviews share three limitations that the present study addresses. First, no prior review covers all four design-relevant dimensions: security architecture, financial stability, privacy and regulatory compliance, and user adoption, within a single integrated protocol. Second, no review applies PRISMA 2020 guidelines to the CBDC literature, making study selection, inclusion criteria, and quality assessment explicit and reproducible. Third, no review proposes a theoretical synthesis framework derived from cross-dimensional evidence. The present study fills these gaps by conducting a PRISMA 2020-compliant systematic review of 140 articles drawn from Web of Science, with a search cutoff of February 2026, and by deriving the CBDC design trilemma as an organizing synthesis of the cross-theme findings.

The remainder of this paper is organized as follows. Section 2 presents the background and theoretical framework. Section 3 describes the systematic review methodology. Section 4 presents the results, including descriptive statistics and thematic findings. Section 5 discusses cross-theme synthesis, answers the research questions, and identifies research gaps. Section 6 concludes with implications and future research directions.

2. Background and Theoretical Framework

2.1. Defining CBDC

A central bank digital currency is a digital form of central bank money that is denominated in the national unit of account and represents a direct liability of the central bank

(BIS, 2020). CBDCs are distinct from commercial bank deposits (which are liabilities of private banks), cryptocurrencies (which lack a central issuer), and stablecoins (which are issued by private entities and backed by reserves).

2.2. CBDC Design Dimensions

The literature identifies several key design dimensions along which CBDCs can vary:

1. Type: Retail (general public) versus wholesale (financial institutions) versus hybrid;
2. Access model: Account-based (identity-verified) versus token-based (bearer instrument) versus hybrid;
3. Architecture: Direct/one-tier (central bank manages all accounts) versus two-tier/intermediated (commercial banks distribute CBDC);
4. Technology: Centralized database versus distributed ledger technology (DLT)/blockchain;
5. Remuneration: Interest-bearing versus non-interest-bearing;
6. Privacy: Full anonymity, pseudonymous, tiered (graduated by transaction value), or fully transparent;
7. Programmability: Basic payment functionality versus smart contract capabilities;
8. Offline capability: Online-only versus offline transaction support.

2.3. The CBDC Design Pyramid

Auer and Böhme (2020) provide a foundational framework for CBDC design through the “CBDC pyramid,” which links consumer needs to key technological and policy design choices (Auer & Böhme, 2020). The pyramid establishes a structured hierarchy: at the base, central banks must decide on the operational architecture (direct vs. intermediated); successive layers address the infrastructure (centralized vs. distributed ledger), access model (account-based vs. token-based), and degree of interlinkage with existing payment systems. As Auer and Böhme (2020) highlight, “a trusted and widely usable retail CBDC must be secure and accessible, offer cash-like convenience and safeguard privacy,” though “various technical designs satisfy these criteria to different degrees, and the associated trade-offs need to be identified.” The pyramid framework underscores that no single architecture fully addresses all objectives, motivating the multi-dimensional analysis in this review.

2.4. Privacy as a Monetary Attribute

Foundational monetary theory establishes privacy as integral to money’s function. Kahn et al. (2005) argue that money serves as a “primitive form of memory,” enabling transactions without identity disclosure (Kahn et al., 2005), while experimental evidence identifies privacy as a third attribute of money alongside liquidity and store of value (Borgonovo et al., 2021). Recent scholarship further reframes payment privacy as a public good: non-anonymous transactions generate negative informational externalities as firms use disclosed data for price discrimination (Garratt & van Oordt, 2021; Kang, 2024). These theoretical foundations that reducing anonymity modifies the utility structure of money itself underpin the privacy–compliance tensions analysed in detail in Themes 3 and 4 (Sections 4 and 5).

2.5. Theoretical Framework: The CBDC Design Trilemma

Drawing on the foundational literature reviewed above, this study proposes a CBDC design trilemma as its central analytical framework. The trilemma holds that central banks face a structural constraint among three objectives:

1. Privacy/Anonymity: The degree to which CBDC transactions can be conducted without revealing the identity or behaviour of the transacting parties. In this framework,

- “privacy” ranges from full anonymity (no party, including the central bank, can link a transaction to an identity) through pseudonymity to full transparency. The relevant measure is the information disclosed per transaction to the issuer and intermediaries.
2. Financial Stability: The preservation of banking sector intermediation, prevention of rapid deposit migration (bank runs), and maintenance of effective monetary policy transmission. The relevant measures are the magnitude of deposit outflows from commercial banks, the speed at which CBDC can substitute for deposits during stress, and the impact on credit supply.
 3. Regulatory Compliance: The ability of authorities to enforce Anti-Money Laundering (AML), Combating the Financing of Terrorism (CFT), Know-Your-Customer (KYC), and tax transparency requirements through the CBDC system. The relevant measure is the degree of transaction traceability and identity verification built into the design.

The trilemma is proposed here as a literature synthesis framework, a structured lens for organizing and interpreting the patterns that emerge from 140 coded articles, rather than a formal impossibility theorem derived from first principles. It does not claim mathematical proof that all three objectives are mutually exclusive; instead, it posits that the existing body of research, when read across themes, reveals a recurring pattern of pairwise tensions that no current or proposed design has simultaneously resolved. The value of the framework lies in making these tensions explicit and testable. Specifically, the trilemma would be falsified if a CBDC design were identified, whether in the reviewed literature or in future deployments, that simultaneously achieves high privacy (anonymity comparable to cash for all transaction sizes), strong financial stability (no measurable increase in deposit migration or run risk), and full regulatory compliance (AML/CFT enforcement equivalent to or exceeding current banking standards). The absence of such a design in the 140-article sample is consistent with the trilemma, but the absence of evidence is not evidence of impossibility. It is equally important to acknowledge that the review maps what the literature reports, not the underlying causal structure of CBDC trade-offs; the patterns identified here may reflect shared assumptions among researchers as much as real-world constraints.

It should be noted that the three vertices are not conceptually symmetric: privacy and financial stability are outcome objectives that CBDC design can affect, while regulatory compliance is an institutional constraint imposed externally by legal frameworks. This asymmetry means that compliance is not “chosen” in the same way that privacy or stability levels are calibrated; it is a binding requirement that shapes the feasible design space from outside. The trilemma is therefore better understood as, given the constraint of compliance, how much privacy and stability can be jointly achieved? An alternative formulation would model this as a constrained optimization problem, maximizing some combination of privacy and stability subject to a minimum compliance constraint rather than a three-vertex trilemma. This formulation is arguably more precise, but the trilemma visualization is retained here for two reasons: (1) it communicates the trade-off structure more intuitively to policymakers, and (2) the degree of compliance enforcement is itself a variable across jurisdictions (strict vs. risk-based AML, FATF grey-listing thresholds), making compliance not a fixed constraint but a movable one, which is why it functions as a vertex that central banks navigate toward or away from, rather than a fixed boundary.

The proposition is that maximizing any two of these objectives necessarily constrains the third: high privacy with strong stability safeguards (holding limits, identity-linked accounts) undermines anonymity; high privacy with regulatory compliance requires a degree of traceability that erodes anonymity by design; and strong stability measures combined with full compliance create frictions that reduce adoption appeal. If this trilemma

holds, it implies that all feasible CBDC designs occupy a “zone of compromise” rather than an optimum, and that the optimal position within this zone varies by jurisdictional context.

To evaluate this proposition against the available evidence, the review organizes the literature into four thematic dimensions that map onto the trilemma’s vertices and linkages:

1. Theme 1: Design Frameworks and Architecture: The design space within which the trilemma operates shows that architectural choices (access model, technology, remuneration) constrain which positions in the trilemma are feasible.
2. Theme 2: Financial Stability, Banking and Liquidity Risk: The stability vertex with evidence on how CBDC affects bank intermediation, monetary policy, and systemic risk.
3. Theme 3: Privacy, Security, and Information Trade-Offs: The privacy–compliance edge evidence on whether and how privacy and regulatory transparency can be reconciled.
4. Theme 4: User Adoption, Trust, and Institutional Quality: The demand-side validation evidence on whether adoption outcomes confirm or contradict the trilemma’s predicted trade-offs.

These four themes are interconnected: design choices (T1) directly influence stability outcomes (T2) and privacy properties (T3), which in turn shape adoption behaviour (T4). Note that the trilemma’s three vertices (privacy, stability, compliance) do not map one-to-one onto the PRQ’s four outcome dimensions (adoption, privacy, inclusion, stability). Rather, adoption and financial inclusion function as downstream consequences of the trilemma: they are shaped by how central banks position themselves within the trilemma’s constraint space. Theme 4, therefore, serves not as a trilemma vertex but as the demand-side test of whether the trilemma’s predicted trade-offs are visible in real-world outcomes.

The systematic review in Section 4 presents the evidence within each theme; the cross-theme synthesis in Section 5 then evaluates the trilemma proposition against the coded data.

3. Materials and Methods

This study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021). The review protocol was designed prior to execution and covers search strategy, selection criteria, data extraction, and quality assessment.

It is important to clarify the analytical approach used in this review. The term “systematic review” in PRISMA 2020 encompasses a range of synthesis methods, from qualitative narrative synthesis to quantitative statistical meta-analysis. The present study employs descriptive quantitative synthesis, specifically frequency analysis and cross-tabulation of coded study characteristics (Section 4.2), rather than statistical meta-analysis, which pools standardized effect sizes across studies to produce a single summary estimate. Statistical pooling is not appropriate here for two reasons. First, the included studies are methodologically heterogeneous: 59% are theoretical models, while the remainder span surveys, empirical analyses, and simulation studies, each reporting different outcome measures that cannot be combined on a common scale. Second, CBDC design, adoption, and stability effects are context-dependent, varying by jurisdiction, CBDC type, and design architecture, making aggregation across studies substantively misleading. The quantitative analysis in this review therefore reports patterns and proportions in the literature rather than pooled effect estimates, and all conclusions should be interpreted accordingly.

3.1. Search Strategy

This review employs a two-tier data structure. The core systematic sample (140 articles) was identified exclusively through the Web of Science (WOS) Core Collection following

a pre-defined, reproducible search protocol. The supplementary corpus (24 additional sources) was identified non-systematically through backward citation tracking and targeted searches of Scopus and SSRN; these sources provide theoretical grounding, methodological context, and case study evidence but are not included in the PRISMA flow count or the quantitative cross-tabulations. This distinction is important: claims about literature-wide patterns rest on the systematic sample; supplementary sources are used only for contextual framing.

WOS was selected as the primary database for three reasons: (1) its coverage of high-impact journals indexed in SCIE and SSCI ensures a quality floor; (2) its structured metadata (accession numbers, citation counts) enables reproducible deduplication; and (3) its cross-disciplinary scope captures CBDC research spanning economics, finance, computer science, and information systems. Scopus and SSRN were used as supplements because CBDC research is heavily influenced by central bank working papers (BIS, ECB, IMF) and policy pre-prints that often precede journal publication; excluding these sources entirely would miss foundational contributions. However, this strategy may under-represent grey literature, conference proceedings, and non-English scholarship, a limitation discussed in Section 6.

Four separate WOS searches were conducted, one per theme, using the following structure: The final searches across all four thematic streams were completed on 28 February 2026.

TS = ("central bank digital currency" OR "CBDC")

AND TS = (<theme-specific terms>)

Theme-specific search terms are detailed in Table 1.

Table 1. Web of Science search strings by theme.

Theme	Additional Search Terms (Combined with CBDC Base String)
T1	"design" OR "architecture" OR "framework" OR "token-based" OR "account-based" OR "two-tier" OR "distributed ledger" OR "blockchain" OR "DLT" OR "retail CBDC" OR "wholesale CBDC" OR "programmable money" OR "smart contract" OR "interoperability"
T2	"financial stability" OR "bank run" OR "disintermediation" OR "liquidity" OR "monetary policy" OR "banking sector" OR "deposit outflow" OR "bank lending" OR "credit supply" OR "systemic risk" OR "money creation" OR "dollarization"
T3	"privacy" OR "anonymity" OR "security" OR "data protection" OR "surveillance" OR "AML" OR "anti-money laundering" OR "KYC" OR "privacy-preserving" OR "zero-knowledge proof" OR "transparency" OR "cybersecurity" OR "GDPR"
T4	"adoption" OR "user adoption" OR "trust" OR "financial inclusion" OR "financial literacy" OR "unbanked" OR "institutional quality" OR "technology acceptance" OR "TAM" OR "UTAUT" OR "cultural values" OR "perceived risk" OR "willingness to adopt"

3.2. Inclusion and Exclusion Criteria

The inclusion and exclusion criteria applied at each screening stage are summarized in Table 2.

Table 2. Inclusion and exclusion criteria.

Eligibility Criterion	Inclusion Criteria	Exclusion Criteria
Topic	CBDC-focused research	Pure cryptocurrency, blockchain without CBDC, stablecoin-only studies
Time range	2014–2026 (inclusive)	Before 2014
Language	English	Non-English
Document type	Article; Review Article	Conference papers, editorials, book chapters
Database index	SCIE, SSCI	Non-indexed journals

3.3. Selection Process (PRISMA Flow)

Figure 1 presents the PRISMA flow diagram. A total of 200 records were retrieved (50 per theme). After removing 54 duplicates (identified by WOS accession number), 146 unique records remained. Title and abstract screening excluded 6 records that were not substantively focused on CBDC (pure cryptocurrency mining, generic blockchain scalability, or stablecoin-only studies without CBDC analysis). The remaining 140 articles were retrieved in full text; all 140 met the inclusion criteria upon full-text assessment. The high inclusion rate (96%) warrants explanation. It reflects a deliberate methodological choice: the search strings required the co-occurrence of “CBDC” or “central bank digital currency” with theme-specific terms (e.g., “disintermediation,” “privacy-preserving,” “adoption”), effectively pre-filtering the corpus at the query stage. The 6 excluded records were articles that appeared in results due to keyword co-occurrence but addressed cryptocurrency mining, generic blockchain scalability, or stablecoin regulation without substantive CBDC analysis. A less restrictive search strategy (e.g., using “digital currency” as the base term) would have yielded more records but also substantially more irrelevant results, requiring a larger exclusion count at screening. The trade-off is that the strict search strings may have missed articles that discuss CBDC-relevant concepts without using the exact term “CBDC”, for instance, papers on “digital money” or “e-money” that pre-date the CBDC terminology. No additional articles were excluded at the full-text stage.

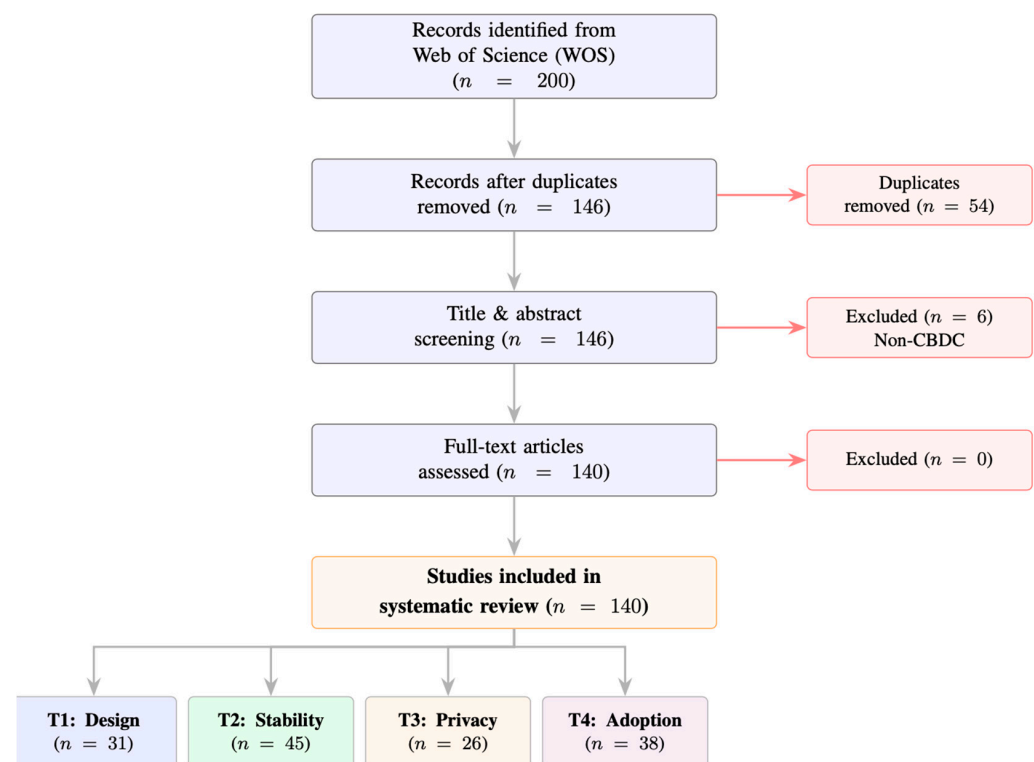


Figure 1. PRISMA 2020 flow diagram. A total of 200 records were retrieved from Web of Science (50 per theme). After removing 54 duplicates, 146 unique records were screened. Six non-CBDC records were excluded, yielding 140 articles included in the systematic review across four thematic dimensions.

3.4. Data Extraction and Coding

Each included article was coded using a structured extraction form with 16 fields (Table 3). Coding was performed based on article abstracts, introductions, and key findings sections. The 140 systematically identified articles form the core analytical sample of this review. In addition, 24 supplementary references are cited throughout the paper to provide

theoretical grounding (e.g., foundational monetary theory, privacy as a public good), methodological context, and comparative case study evidence. These supplementary sources drawn from Scopus, SSRN, and institutional working paper series (BIS, ECB, central bank staff papers) were identified through backward citation tracking and targeted searches. They are not included in the PRISMA flow count but are clearly distinguished in the bibliography.

Table 3. Coding framework fields.

Field	Response Categories/Description
Article ID	Unique identifier (e.g., T1-001)
Authors, Year, Journal	Bibliographic metadata
Country/Jurisdiction	Geographic focus of the study
Research Method	Empirical/Theoretical/Mixed/Case study/Survey/Review
CBDC Type	Retail/Wholesale/Both/NA
Design Model	Account-based/Token-based/Hybrid/NA
Architecture	One-tier/Two-tier/Intermediated/NA
Primary Theme	T1/T2/T3/T4
Privacy Approach	Full anonymity/Pseudonymous/Tiered/None/NA
Financial Stability Impact	Positive/Negative/Mixed/NA
Key Findings	Summary of main results
Relevance to PRQ	High/Medium/Low

3.5. Quality Assessment

Each included article was assessed for methodological rigour using four criteria: (1) clarity of research question and objectives, (2) appropriateness of methodology to the stated aims, (3) transparency of data sources and analytical procedures, and (4) strength of evidence supporting the conclusions. Articles were rated as high, medium, or low relevance to the primary research question. Of the 140 articles, 34 (24.3%) were rated high relevance, 72 (51.4%) medium, and 34 (24.3%) low. Low-relevance articles were retained in the sample as they contributed to the descriptive mapping of the field, but thematic synthesis prioritized findings from high- and medium-relevance studies. Because coding was performed by a single researcher, a random subsample of 20 articles (14.3%) was recoded after a two-week interval to assess intra-coder consistency, yielding a Cohen's kappa of 0.81 across key categorical fields (CBDC type, design model, architecture, privacy approach), indicating substantial agreement. Two methodological limitations of the coding process should be noted. First, coding was based primarily on article abstracts, introductions, and key findings sections rather than full-text close reading of all 140 articles. This approach is standard in large-sample SLRs but may miss nuanced positions or qualifications buried in discussion sections. Second, single-researcher coding introduces the risk of systematic bias in interpretive judgements, particularly for fields like "financial stability impact" and "relevance," which require subjective assessment. The results of the coded analysis should therefore be understood as interpretive aggregations that indicate patterns in the literature, not as objective measurements of underlying phenomena.

4. Results

4.1. Descriptive Statistics

4.1.1. Temporal Distribution

Although the search window covered 2014–2026, the earliest CBDC-specific article retrieved dates to 2019, reflecting the nascent state of the field before that year. Seven articles classified as 2026 were accepted or published online ahead of print before the search cutoff date and are therefore included. The 140 studies thus span 2019–2026 (Table 4). CBDC research has grown rapidly, with publications increasing from 1 article in 2019 to a peak of 41 in 2024. Crucially, this growth is uneven across themes: T2 (stability) and T4 (adoption) account for 59% of the total and grew fastest, while T3 (privacy) remains the smallest strand (19%). This asymmetry matters for the trilemma: the evidence base for the stability vertex is roughly twice as deep as for the privacy vertex, meaning that the literature’s convergence on stability safeguards rests on a much larger body of work than its privacy design recommendations, a structural imbalance that may bias the dominant recommendations toward over-weighting stability at the expense of privacy.

Table 4. Article distribution by year and theme.

Publication Year	T1: Design Frameworks	T2: Financial Stability	T3: Privacy & Security	T4: User Adoption	Annual Total
2019	0	1	0	0	1
2020	1	1	0	0	2
2021	3	4	0	0	7
2022	5	8	4	6	23
2023	3	8	9	10	30
2024	9	17	6	9	41
2025	8	5	6	10	29
2026	2	1	1	3	7
Total	31	45	26	38	140

4.1.2. Methodological Profile

The literature is predominantly theoretical (59.3%), with DSGE models, general equilibrium frameworks, and mechanism design approaches dominating (Table 5). Empirical studies (17.9%) are growing but remain concentrated on China’s e-CNY pilot, while survey-based research (5.7%) clusters almost entirely in T4 (adoption). This methodological profile has a direct consequence for the trilemma: the stability–privacy edge rests largely on model-derived predictions rather than observed outcomes. T2 (stability) is 73% theoretical, meaning the disintermediation warnings that dominate policy discussions are projections from DSGE models whose behavioural assumptions, particularly about depositor switching behaviour, remain largely untested against real deployment data.

4.1.3. Geographic Focus

The EU/Euro Area (44 articles) and China (42) dominate the geographic focus, followed by the USA (23) and UK (19), where counts reflect the number of articles that substantively discuss each jurisdiction. This reflects the prominence of the Digital Euro investigation and China’s e-CNY pilot as focal points for CBDC research. Developing countries are represented through the Bahamas (14), Nigeria (6), and India (5).

Table 5. Distribution of research methods.

Research Method	Article Count (<i>n</i>)	Share (%)	T1: Design Frameworks	T2: Financial Stability	T3: Privacy & Security	T4: User Adoption
Theoretical/Modelling	83	59.3	13	33	16	21
Empirical	25	17.9	6	7	3	9
Review	11	7.9	6	2	1	2
Mixed Methods	10	7.1	4	3	2	1
Survey	8	5.7	0	0	3	5
Case Study	3	2.1	2	0	1	0
Total	140	100	31	45	26	38

4.2. Quantitative Synthesis of Coded Variables

Beyond the descriptive statistics, cross-tabulation of the coded variables reveals several substantively notable patterns across the 140-article sample. An important caveat applies to all quantitative results in this subsection: the cross-tabulations represent descriptive aggregations of how the literature addresses CBDC design, not causal inferences about CBDC effects. The unit of analysis is the article (what researchers find or recommend), not the CBDC itself. Patterns such as “71.4% recommend tiered privacy” indicate convergence in scholarly opinion, not empirical proof that tiered privacy is optimal. Odds ratios and vote counts summarize the distribution of findings across studies with heterogeneous methods, samples, and assumptions; they should be interpreted as indicative of literature-wide tendencies rather than definitive effect sizes. Results are also sensitive to coding choices: the “stability impact” and “relevance” fields involve interpretive judgement by a single coder, and different classification decisions could shift the distributions reported below. A further boundary condition applies: the sample is overwhelmingly focused on retail CBDC (41 retail-focused, 25 covering both, only 1 wholesale-only). The trilemma framework and all findings derived from it are therefore primarily applicable to retail CBDC design; whether analogous trade-offs bind in wholesale or cross-border CBDC contexts remains an open question.

4.2.1. Stability Assessment by Theme

A vote-count analysis of the 69 articles providing a directional stability assessment suggests that the literature is predominantly cautious: 58.0% report mixed (design-dependent) effects, 27.5% report negative effects, and only 14.5% report unambiguously positive effects. However, this aggregate distribution masks sharp thematic variation (Table 6).

Table 6. Financial stability assessment by theme.

Theme	Negative (%)	Mixed (%)	Positive (%)	<i>n</i>
T1—Design	0 (0)	9 (69)	4 (31)	13
T2—Stability	18 (49)	17 (46)	2 (5)	37
T3—Privacy	0 (0)	3 (60)	2 (40)	5
T4—Adoption	1 (7)	11 (79)	2 (14)	14
Total	19 (27.5)	40 (58.0)	10 (14.5)	69

Theme 2 is the only theme where negative assessments dominate (49%), reflecting the concentration of disintermediation and bank-run modelling in this strand. In contrast, T1 and T3 articles that address stability tend toward mixed or positive assessments, suggesting

that when stability is considered alongside design or privacy objectives, more nuanced and conditional conclusions emerge. This asymmetry implies that the “CBDC threatens banking” narrative is driven primarily by the financial stability sub-literature, while researchers approaching CBDC from design, privacy, or adoption perspectives are more likely to identify mitigating design features.

4.2.2. Research Method and Quality

The relationship between research method and assessed relevance suggests a notable pattern. Empirical and survey-based studies are more than twice as likely to receive a high-relevance rating compared to theoretical studies (odds ratio = 2.31; Table 7). This finding should be treated as exploratory: the relevance ratings were assigned by a single coder, and the observed association may partly reflect a coding bias toward valuing empirical evidence over theoretical modelling, rather than an inherent quality difference.

Table 7. Research method and relevance assessment.

Research Method	High (%)	Medium (%)	Low (%)	<i>n</i>
Theoretical/Modelling	13 (21)	35 (56)	15 (24)	63
Empirical	8 (35)	14 (61)	1 (4)	23
Survey	4 (44)	4 (44)	1 (11)	9

Note: Subsample of articles with relevance ratings assigned.

This disparity carries a pointed implication for the trilemma: the edges that are most consequential for policy, whether holding limits prevent runs and whether tiered privacy satisfies users in practice, are precisely those where the evidence is thinnest. Theoretical models can demonstrate that a trilemma exists in principle, but only empirical studies can establish where the binding constraints bite in specific jurisdictions. The $2.3\times$ relevance advantage of empirical work (Table 7) signals that reviewers and coders independently judged practice-grounded studies as more informative, reinforcing the urgency of the empirical evidence gap identified in Section 5.

4.2.3. Privacy Design Recommendations

Cross-tabulating the privacy approach variable across themes shows a marked tendency toward tiered privacy as the most frequently recommended design (Table 8).

Table 8. Privacy design recommendations by theme (articles specifying a privacy approach).

Privacy Design Approach	T1: Design Frameworks	T2: Financial Stability	T3: Privacy & Security	T4: User Adoption	<i>n</i> (%)
Full anonymity	1	0	1	0	2 (3.6)
Pseudonymous	6	0	2	0	8 (14.3)
Tiered privacy	6	8	12	14	40 (71.4)
None specified	2	0	3	1	6 (10.7)
Total	15	8	18	15	56 (100)

Tiered privacy accounts for 71.4% of all explicit privacy design recommendations across the sample but this convergence is itself indicative of the trilemma at work. Full anonymity (the privacy vertex) is effectively abandoned (3.6%), not because it is technically infeasible but because it is incompatible with the compliance vertex. Conversely, no-privacy designs (the compliance vertex) are recommended by only 10.7%, because they deter adoption. Tiered privacy is not an optimal solution; it is consistent with the interior

compromise the trilemma predicts. However, an alternative explanation must be acknowledged: the 71.4% convergence may also reflect policy bandwagoning (researchers following the ECB’s tiered privacy proposal), institutional conservatism (defaulting to “balanced” recommendations), or citation clustering (later articles adopting the framing of influential early papers). The trilemma interpretation and the policy-diffusion interpretation are not mutually exclusive, but the data cannot distinguish between them. The unanimity in T4 where all 14 privacy-coded articles recommend tiered approaches suggests that adoption researchers, who are closest to user-facing evidence, view this compromise as the only design that can survive contact with actual users.

4.2.4. CBDC Type and Stability Outcomes

Among articles specifying CBDC type and providing a stability assessment ($n = 41$), retail-focused studies report positive or mixed effects at a higher rate (74%) than articles examining both retail and wholesale variants (64%). This 10-percentage-point gap, while based on a small wholesale subsample, points to a blind spot in the trilemma framework as currently constructed: the trilemma’s vertices are calibrated to retail CBDC, yet wholesale dimensions, such as cross-border settlement, interbank liquidity re-allocation, and currency substitution, may introduce a separate set of stability–compliance tensions that the retail-focused literature does not capture. The near-total absence of wholesale-only studies ($n = 1$) means that one entire class of CBDC design operates outside the evidence base on which the trilemma rests. The results are summarized in Table 9.

Table 9. Stability assessment by CBDC type.

CBDC Type	Negative (%)	Mixed (%)	Positive (%)	<i>n</i>
Retail	7 (26)	15 (56)	5 (19)	27
Both (retail + wholesale)	5 (36)	7 (50)	2 (14)	14

These cross-tabulations establish the quantitative scaffolding against which the thematic findings below should be read: the trilemma is not merely asserted but grounded in measurable patterns of convergence and divergence across 140 coded articles.

4.3. Theme 1: CBDC Design Frameworks and Architecture

The 31 articles in Theme 1 define the design space within which the trilemma operates. A fundamental finding is that no system can simultaneously achieve universal access, perfect security, and full privacy (Kahn et al., 2018). Each architectural choice (access model, technology, remuneration) determines which positions along the privacy–stability–compliance triangle are feasible, and which are foreclosed.

4.3.1. Access Model: Token-Based Versus Account-Based

No single access model dominates the literature. Of the 14 T1 articles that explicitly recommend an access model, hybrid designs are the most frequently proposed ($n = 7$), followed by token-based ($n = 4$) and account-based ($n = 3$). The distinction lies in verification logic: token-based systems verify the payment instrument itself (analogous to physical cash), offering stronger privacy but posing challenges for regulatory control; account-based systems verify the holder’s identity, enhancing traceability and AML compliance but reducing privacy (Kahn et al., 2018). Hybrid designs are favoured because they balance the privacy benefits of tokens with the regulatory compliance advantages of accounts (Lamberty et al., 2024; Ok et al., 2025; J. Zhang et al., 2021). Notably, even when CBDCs emulate features of cash (e.g., offline use, peer-to-peer transfer), they will likely include some form of traceability, making them “account-based systems with token-like properties”

rather than pure digital cash (Auer & Böhme, 2020). A legal strand of this debate argues that CBDC should replicate the legal characteristics of physical banknotes through token-based design (Geva et al., 2021), while a competing view holds that tokenization is conceptually misapplied and that CBDC must be understood as a direct claim on the central bank regardless of access model (Milne, 2023). Where token-based designs are advocated, improved public IT literacy is identified as a precondition for their viability (Stroukal & Peterka, 2025).

4.3.2. Architecture: One-Tier Versus Two-Tier

A near-consensus favours two-tier (intermediated) architecture, where the central bank issues CBDC but delegates distribution and customer-facing services to commercial banks or payment service providers. Of 31 T1 articles, 22 recommend or assume a two-tier architecture. The rationale is threefold: (1) it preserves the role of commercial banks in credit intermediation, reducing disintermediation risk (Opore & Kim, 2020); (2) it leverages existing banking infrastructure rather than requiring central banks to build retail-facing systems (Grosman et al., 2024); and (3) it aligns with the BIS foundational principles (BIS, 2020). One-tier (direct) architecture is discussed but rarely recommended; only two articles advocate for it, both in the context of promoting financial inclusion in underbanked jurisdictions where commercial banking infrastructure is inadequate (Didenko & Buckley, 2022).

4.3.3. Technology: DLT Versus Centralized

The technology debate has evolved significantly. Early articles (2019–2021) frequently assumed blockchain or DLT as the default CBDC infrastructure, but more recent scholarship is skeptical of DLT's suitability for national-scale payment systems due to throughput limitations, energy costs, and governance complexity (Sethaput & Innet, 2023; Jin & Xia, 2022). The emerging consensus distinguishes between retail CBDC (where centralized databases are more efficient) and wholesale CBDC (where DLT offers advantages for cross-border settlement and atomic transactions). Hybrid approaches with centralized core systems and DLT-based subsystems for specific functions (e.g., offline payments, cross-border settlement) are increasingly proposed (Tunzina et al., 2024; Atangana et al., 2026).

4.3.4. Remuneration: Interest-Bearing Versus Non-Interest

This design dimension sits directly on the stability–adoption edge of the trilemma. Interest-bearing CBDC enhances its attractiveness as a savings vehicle and strengthens monetary policy transmission (the interest rate on CBDC provides a new policy tool) but directly competes with bank deposits and amplifies disintermediation risk (Agur et al., 2022; Keister & Sanches, 2022). Non-interest-bearing CBDC reduces competitive pressure on banks but limits its policy utility and adoption appeal. The literature's modal recommendation is zero remuneration, reflecting the dominance of stability concerns over adoption objectives. Parra-Polania and Martinez-Ventura (2026) formalize this trade-off, showing that the optimal interest rate depends on the central bank's relative weighting of privacy and financial inclusion objectives.

4.3.5. Cross-Border and Interoperability

A small but growing strand (eight articles) addresses cross-border CBDC design. The key challenge is interoperability: how different national CBDCs can interact for cross-border payments without compromising domestic monetary sovereignty (Kuehnlenz et al., 2022; Moro & Nispi Landi, 2024). Multi-CBDC platforms (e.g., mBridge) are identified as promising but raise complex questions about governance, exchange rate management, and regulatory harmonization. The cross-border dimension is important for the trilemma

because it introduces a fourth potential vertex, monetary sovereignty, that may be in tension with interoperability, particularly for smaller or developing economies (Ferrari Minesso et al., 2022).

Taken together, the 31 Theme 1 articles define not a single optimal CBDC design but a feasibility boundary: the architectural choices made at this level determine which positions along the trilemma are available to subsequent design decisions on privacy and compliance. The near-consensus for two-tier, non-interest-bearing architecture is best understood as a collective choice by the literature to protect the stability vertex, preserving bank intermediation and limiting disintermediation risk at the cost of constraining the privacy and utility objectives that would attract users. This is not a solution to the trilemma but a prioritization within it. The theoretical implication for CBDC design is that architecture is a pre-commitment: once a two-tier, account-based, non-interest-bearing structure is adopted, the range of achievable privacy and compliance outcomes narrows substantially. Policymakers should therefore treat architecture selection as the primary trilemma decision, recognizing that downstream choices on anonymity levels and AML controls are constrained by the architectural foundation chosen at this stage.

4.4. Theme 2: Financial Stability, Banking, and Liquidity Risk

The 45 articles in Theme 2 form the largest and most theoretically developed strand. The central concern is disintermediation: if CBDC is sufficiently attractive, depositors may shift funds from commercial banks to the central bank, reducing the deposit base that funds bank lending and potentially triggering bank runs during financial stress.

4.4.1. Disintermediation Risk

The disintermediation literature spans a wide range of predicted severity. At one extreme, models predict that CBDC could cause “significant deposit outflows” and “credit crunches” if introduced without safeguards (Fernández-Villaverde et al., 2020). At the other, the Brunnermeier–Niepelt equivalence result argues that under certain conditions (central bank lending to banks replacing lost deposits), CBDC introduction has no real effects on banking (Niepelt, 2024). The majority position is conditional: disintermediation is a real risk but can be managed through design safeguards such as holding limits, zero remuneration, and two-tier distribution. Y. S. Kim and Kwon (2022) demonstrate that CBDC can even *increase* bank lending if the central bank recycles CBDC deposits back to commercial banks, while Schilling et al. (2024) show that the effect depends critically on whether the central bank commits to a fixed or flexible CBDC supply.

4.4.2. Bank Run Dynamics

A distinctive contribution of the T2 literature is the analysis of CBDC as a “run accelerator.” Digital bank runs can propagate faster than traditional runs because CBDC enables instant, frictionless conversion of deposits to central bank money without physical withdrawal constraints (Sanchez-Roger & Puyol-Antón, 2021). Bitter (2024) models CBDC-enabled runs and finds that holding limits are effective but must be set sufficiently low to prevent “dash for safety” dynamics during crises. The critical insight is that the existence of CBDC as an alternative safe asset changes depositor behaviour, even if the CBDC is not actively used in normal times, a “latent run risk” that affects bank funding costs and credit pricing.

4.4.3. Monetary Policy Transmission

CBDC’s effects on monetary policy are contested. Proponents argue that interest-bearing CBDC provides a direct transmission channel to households, bypassing the banking sector and reducing the zero-lower-bound constraint (Barrdear & Kumhof, 2021). The

negative-interest-rate capability of CBDC is particularly emphasized: during deflationary episodes, CBDC could enable negative rates on household balances that are impossible with physical cash (Hua et al., 2025). Critics counter that CBDC monetary policy effectiveness depends on adoption rates and substitution patterns that are currently unknown (Abad et al., 2025), and that the distributional effects (penalizing savers) may undermine public trust (Pfister, 2024). Park (2024) demonstrates that in economies with large excess reserves, CBDC's monetary policy impact is diminished because reserves already satisfy liquidity demand.

4.4.4. Empirical Evidence on Stability

The empirical sub-strand within T2 is small but growing. Heitmann et al. (2025) provide the first cross-country panel analysis of CBDC effects on bank stability, finding that CBDC adoption stages (research, pilot, launch) are associated with modest but statistically significant reductions in bank profitability. Xiao et al. (2024) use China's e-CNY pilot as a natural experiment and find that CBDC introduction is associated with higher corporate debt financing costs, suggesting that the transparency effects of CBDC extend beyond individual transactions to corporate financial behaviour. Beckmann et al. (2024) analyse US bank stock market reactions to CBDC announcements and find negative abnormal returns, indicating that investors perceive CBDC as a threat to bank franchise value.

The Theme 2 literature collectively operationalizes financial stability not as an objective to be maximized but as a floor to be defended. The dominant policy recommendation on interest-bearing, quantity-capped CBDC distributed through a two-tier intermediary structure emerges from minimizing the threat to bank intermediation rather than from optimizing CBDC capabilities. The 49% of T2 articles reporting negative or mixed stability assessments, and the growing empirical evidence that CBDC adoption stages are associated with reductions in bank profitability (Heitmann et al., 2025) and higher corporate debt financing costs (Xiao et al., 2024), confirm that the stability concerns identified in theoretical models are not hypothetical. The theoretical implication is that stability protection dominates the design agenda in this strand: the literature has settled on a stability-first design principle that accepts reduced adoption appeal and limited monetary policy innovation as the price of bank system preservation. For policymakers, this means that any CBDC design that genuinely improves on the current non-interest-bearing benchmark by offering interest, programmability, or greater anonymity will require an explicit justification for why the associated increase in disintermediation risk is acceptable in the specific jurisdictional context.

4.5. Theme 3: Privacy, Security, and Information Trade-Offs

The 26 articles in Theme 3 address the privacy–compliance edge of the trilemma, arguably the most technically complex and politically sensitive dimension.

4.5.1. The Privacy–Compliance Spectrum

The literature conceptualizes privacy along a spectrum from full anonymity (no party can link transactions to identities) through pseudonymity (identifiers exist but are not directly linked to real identities) and tiered/controllable anonymity (different privacy levels for different transaction sizes or contexts) to full transparency (all transactions are traceable by authorities). A foundational finding is that “full anonymity is inconsistent with financial integrity standards” (Pocher & Veneris, 2021). FATF requirements effectively rule out the anonymity endpoint. Conversely, full transparency creates “Orwellian payment systems” that deter adoption and may violate fundamental rights to privacy (Westermeier, 2022; Tsang et al., 2023).

4.5.2. Privacy-Preserving Technologies

The technical privacy literature focuses on cryptographic approaches that can shift the privacy–compliance frontier. Key technologies include: (1) zero-knowledge proofs (ZKPs), which allow a party to prove that a transaction satisfies regulatory conditions without revealing transaction details (Ok et al., 2025; Michalopoulos et al., 2025); (2) blind signatures, which allow the issuer to validate CBDC tokens without learning the identity of the holder; (3) ring-confidential transactions, which hide transaction amounts within a group of possible values; and (4) architectural separation, where identity management and transaction processing are handled by different entities to prevent any single party from linking the two (Islam & IN, 2022). These technologies are theoretically promising but face practical challenges: ZKPs are computationally expensive and may not achieve the throughput required for national payment systems; blind signatures require trusted setup assumptions; and architectural separation depends on governance structures that prevent collusion between entities (Takaragi et al., 2022).

4.5.3. Offline Privacy

Offline CBDC transactions present a particular privacy–compliance challenge. Offline capability is essential for financial inclusion (serving areas without internet connectivity) and disaster resilience, but offline transactions cannot be verified in real-time against AML databases. Michalopoulos et al. (2025) provide the most comprehensive analysis of offline privacy–compliance design options, identifying five architectural approaches with different trade-off profiles. The fundamental tension is that offline privacy requires some form of “stored value” that operates independently of central verification, resembling physical cash and inheriting both its privacy benefits and its regulatory challenges.

4.5.4. Surveillance Concerns

A critical and underexplored dimension is the use of CBDC as a surveillance tool. Westermeier (2022) analyses CBDC within the broader context of financial surveillance regimes, arguing that CBDC represents a qualitative shift from transaction-level monitoring to population-level financial surveillance. Tsang et al. (2023) argue for constitutional constraints on central bank access to CBDC data, proposing that central bank independence should be balanced by “CBDC data independence,” “institutional separation of monetary policy and surveillance functions. The Chinese e-CNY’s “controllable anonymity” model, where transactions are anonymous to counterparties but traceable by authorities under legal warranties, has been analysed as a case study of this tension (Broby, 2024; Yue, 2025). Empirical evidence from Korea shows that privacy design variations have significant effects on stated adoption intentions, with stronger effects for privacy-sensitive transaction types (Choi et al., 2024).

The Theme 3 evidence identifies the privacy–compliance edge as the most technically constrained edge of the trilemma. The convergence of 71.4% of privacy design recommendations on tiered privacy represents a collective acknowledgement across the literature that full anonymity and full regulatory compliance are mutually exclusive under current technology and legal frameworks: the field has settled on a middle position without resolving the underlying tension. The 54% of T3 articles identifying an irreconcilable conflict between anonymity and AML/CFT enforcement confirms that this is not a gap addressable by better engineering alone; it reflects a structural conflict between the informational requirements of financial surveillance and the informational minimization that privacy demands. The theoretical implication is that resolving this edge of the trilemma requires not only technical innovation such as privacy-preserving cryptographic protocols but institutional innovation

in the form of legal frameworks that define and enforce the boundary between monetary privacy and lawful surveillance.

4.6. Theme 4: User Adoption, Trust, and Institutional Quality

The 38 articles in Theme 4 provide demand-side evidence that tests whether the trilemma's predicted trade-offs are visible in user behaviour and institutional outcomes.

4.6.1. Adoption Drivers and Barriers

Across 38 T4 articles, five factors consistently emerge as the strongest adoption drivers: (1) trust in the issuing institution (Chua et al., 2025; Osakwe et al., 2024), (2) perceived usefulness relative to existing payment methods (Liu et al., 2025; Sha et al., 2024), (3) financial literacy (Mertzanis et al., 2025; Wu et al., 2024), (4) financial inclusion motivation (particularly in developing countries) (Didenko & Buckley, 2022; Hoang et al., 2023), and (5) social influence and peer effects (Bai et al., 2025; Ma et al., 2022). Five corresponding barriers are identified: (1) privacy and surveillance concerns (Tronnier et al., 2022; Abramova et al., 2022; Jabbar et al., 2023), (2) perceived risk (security, operational, financial) (J. J. Kim et al., 2023; Radic et al., 2022), (3) switching costs from existing payment methods (Desai, 2024; Son et al., 2022), (4) lack of awareness and understanding (Umar, 2025), and (5) digital divides (age, income, geographic) (Bijlsma et al., 2023; Yang et al., 2025).

4.6.2. The Trust–Institutional Quality Nexus

A distinctive T4 finding is the critical role of institutional quality in mediating adoption. Hoang et al. (2023) provide cross-country evidence that government effectiveness is positively associated with public support for CBDC, but—and this is the critical nuance—the relationship is non-linear. Very high institutional quality is associated with *slower* CBDC development, not faster. Tunc (2026) formalizes this as a “regulatory paradox”: countries with the strongest regulatory frameworks (typically advanced economies) progress most slowly in CBDC implementation because they internalize the compliance costs and institutional risks that less regulated jurisdictions externalize. This finding challenges the assumption that strong institutions straightforwardly facilitate CBDC adoption and suggests a deeper tension between institutional quality and innovation speed. Umar (2025) examines CBDC adoption stages across countries and finds that institutional determinants vary by stage: research initiation is driven by technological readiness, while pilot launches are driven by financial inclusion needs and competitive pressure from private digital currencies.

4.6.3. Advanced- Versus Developing-Economy Dynamics

The adoption literature reveals a sharp divide between advanced- and developing-economy contexts. In advanced economies (EU, Japan, Australia), the primary challenge is the “demand gap” (Demertzis & Martins, 2023): consumers already have efficient, trusted payment systems, and the value proposition of CBDC is unclear (Li, 2022; Fujiki, 2023; Bijlsma et al., 2023). Li (2022) estimates that only 4–16% of US consumers would use CBDC, concentrated among the unbanked and privacy-conscious. In developing economies (Nigeria, India, Pacific Islands), CBDC is framed as a financial inclusion tool with clearer demand, but adoption is constrained by infrastructure, literacy, and trust barriers (Osakwe et al., 2024; Didenko & Buckley, 2022; Sandhu et al., 2023).

4.6.4. Comparative Case Studies

The review synthesizes evidence from four CBDC deployments at various stages (Table 10).

Table 10. Comparative case study summary of CBDC deployments.

Jurisdiction & System Name	Architecture & Design Features	Primary Policy Motivation	Adoption Outcome (Source)	Principal Lesson Learned
China (e-CNY)	Two-tier, non-interest, controllable anonymity	Payment efficiency; reduce private platform risk	Low; concentrated in trial periods (Bai et al., 2025)	Government promotion insufficient without merchant uptake
Nigeria (eNaira)	Account-based, two-tier	Financial inclusion; remittance efficiency	<2% active usage (Osakwe et al., 2024)	Trust, infrastructure, and literacy are preconditions
Bahamas (Sand Dollar)	Two-tier, token-based	Geographic inclusion; disaster resilience	Minimal relative to cash (BIS, 2020)	Innovation alone does not drive adoption
Eurozone (Digital Euro)	Investigation phase; tiered privacy proposed	Monetary sovereignty; privacy	N/A—not yet launched	Demand gap in advanced economies (Tronnier et al., 2022 ; Demertzis & Martins, 2023)

Three cross-case patterns emerge. First, all four deployments or proposals adopt two-tier architectures, consistent with the dominant recommendation identified in Theme 1. Second, adoption outcomes have been uniformly disappointing where CBDCs have launched: China’s e-CNY remains “low and concentrated primarily in the trial period” despite strong government intervention ([Bai et al., 2025](#)), Nigeria has achieved less than 2% active usage ([Osakwe et al., 2024](#)), and the Bahamas’ Sand Dollar is “almost nonexistent compared with cash” ([BIS, 2020](#)). Third, the binding constraint differs by development level: EMDEs face infrastructure and trust barriers, while advanced economies face a demand gap; consumers already have efficient payment options, and the value-added case for CBDC remains contested. Corporate-level effects in China (reduced cash holdings, higher debt financing costs from increased transparency ([G. Zhang & Chan, 2025](#); [Xiao et al., 2024](#)) suggest that even where individual adoption lags, CBDC may reshape corporate financial behaviour.

5. Discussion

The evidence from the four thematic strands reviewed in Section 4 collectively grounds the trilemma proposed in Section 2.5 in the systematic literature rather than in a priori reasoning. Theme 1 (31 articles on design architecture) establishes the structural foundation: the literature converges on a finding that no CBDC system can simultaneously achieve universal access, full security, and complete privacy, and that each architectural choice forecloses certain positions along the privacy–stability–compliance triangle. Theme 2 (45 articles on financial stability) supplies the evidence for the stability vertex: the dominant policy recommendation on interest-bearing, quantity-capped CBDC is precisely designed to limit the privacy and utility that would maximize the other two vertices. Theme 3 (26 articles on privacy and compliance) provides the sharpest evidence for the trilemma: 14 articles (54%) identify a direct and irreconcilable tension between anonymity and AML/CFT enforcement, and tiered privacy accounts for 71.4% of all explicit design recommendations precisely because no design achieves both objectives fully. Theme 4 (38 articles on adoption and trust) closes the loop from the demand side: the same compliance and stability-protective features that the literature recommends are identified as barriers to adoption, confirming that design choices that strengthen two vertices of the

trilemma consistently weaken the third. Taken together, the four themes do not each independently prove the trilemma; rather, they converge on a consistent pattern across 140 studies in which the three objectives are never simultaneously maximized. Section 5.1 evaluates each pairwise edge in detail.

5.1. Evaluating the CBDC Design Trilemma

This paper proposes in Section 2 that central banks face a structural design trilemma between privacy, financial stability, and regulatory compliance. The thematic and quantitative evidence presented in Section 4 allows us to evaluate each edge of this proposition. This section argues that the evidence, while predominantly theoretical and not yet validated by live-deployment data, is broadly consistent with all three edges, though with important asymmetries in evidential strength.

5.1.1. Privacy Versus Compliance: Supported

The literature aligns most clearly with this edge. Of 26 T3 articles, 14 (54%) identify a direct and irreconcilable conflict between anonymity and AML/CFT enforcement. Compliance requirements effectively “rule out truly anonymous payments” (Choi et al., 2024), while full transparency deters adoption among privacy-sensitive users (Abramova et al., 2022; Bijlsma et al., 2023). The quantitative synthesis reinforces this: full anonymity is recommended by only 2 of 56 privacy-coded articles (3.6%), while tiered privacy, a compromise that partially sacrifices both privacy and compliance, accounts for 71.4%. No article in the sample proposes a design that achieves both full anonymity and full regulatory compliance.

5.1.2. Privacy Versus Stability: Supported

An anonymous CBDC is harder to cap or monitor, amplifying run risk. Of 45 T2 articles, 17 (38%) recommend holding limits or identity-linked accounts that necessarily reduce anonymity (Fegatelli, 2022; Parra-Polania & Martinez-Ventura, 2026). The cross-tabulation of architecture and stability assessment shows that stability-protective features (holding limits, zero interest, two-tier distribution) are precisely those that constrain CBDC utility and anonymity. The stability literature’s dominant recommendation on interest-bearing, quantity-capped CBDC distributed through intermediaries is a direct sacrifice of the privacy vertex to protect the stability vertex.

5.1.3. Stability Versus Compliance: Partially Supported

This edge is less binary. Stringent compliance (identity verification, transaction monitoring) increases operational friction and reduces adoption, which in turn limits the policy benefits that motivated CBDC issuance (Tunc, 2026; Osakwe et al., 2024). The “regulatory paradox” (Section 4) provides macro-level evidence: countries with the strongest compliance frameworks progress most slowly in CBDC development, suggesting that compliance and effective deployment exist in tension. However, the T4 evidence shows that this tension can be partially moderated by institutional trust in high-trust jurisdictions; users accept compliance requirements more readily (Chua et al., 2025).

5.1.4. The Zone of Compromise

Since the evidence is broadly consistent with all three pairwise trade-offs, the literature suggests that no feasible CBDC design can fully maximize all three objectives. Two epistemological clarifications are necessary here.

First, this review’s object of study is the literature on CBDC design, not CBDC design itself. The trilemma is a pattern observed in what researchers report, recommend, and model, not a direct observation of how real-world CBDCs perform. The distinction matters:

the literature is dominated by theoretical models (59.3%) whose predictions about trade-offs depend on assumptions (e.g., high deposit CBDC substitutability) that remain largely untested against live-deployment data. The trilemma may therefore describe the structure of scholarly reasoning about CBDCs as much as the structure of CBDCs themselves. Future empirical work from live deployments will be essential to determine whether the trade-offs identified in the literature materialize in practice or prove less binding than models predict.

Second, this paper uses the term “structural constraint,” but the trilemma’s status is observational, not deductive. Unlike the Mundell–Fleming trilemma (derived from arbitrage conditions) or Arrow’s impossibility theorem (proven mathematically), the CBDC design trilemma is an empirical pattern: no existing or proposed design in the 140 reviewed articles simultaneously achieves high performance on all three dimensions. The constraint is not logically absolute but practically binding given current technology and regulatory norms analogous to the “blockchain trilemma” (T. Zhang & Huang, 2021), which similarly describes an observed engineering constraint rather than a mathematical impossibility.

This distinction matters because it raises a second question: are the observed trade-offs true constraints inherent in the problem, or merely reflections of current policy preferences? The answer, this paper argues, is both, but in different proportions across the three edges. The privacy–compliance edge is closest to a true constraint: information once disclosed to a counterparty cannot be “undisclosed,” and AML/CFT enforcement inherently requires some degree of traceability that is logically incompatible with full anonymity. The privacy–stability edge is more contingent on policy choice: holding limits and identity-linked accounts are chosen safeguards, not physical laws; a central bank willing to tolerate greater disintermediation risk could relax these constraints. The stability–compliance edge is the most policy-dependent: the “regulatory paradox” (strong regulators moving slowest) reflects institutional caution, not an impossibility. Recognizing this heterogeneity is important: it means the trilemma’s edges are not equally rigid, and policy innovation may loosen some edges more readily than others.

This design trilemma is also distinct from the central banking trilemma formalized by Schilling et al. (2024), which demonstrates that a central bank intermediating deposits cannot simultaneously achieve efficient allocation, financial stability (absence of runs), and price stability. The design trilemma operates at the architectural level (how to build the system), while the central banking trilemma operates at the macroeconomic-policy level (how to manage the consequences). Together, they illustrate that CBDC faces binding constraints at both layers.

5.1.5. Can Technology Dissolve the Trilemma?

The privacy-preserving technologies reviewed in Theme 3, such as zero-knowledge proofs, blind signatures, ring-confidential transactions, and architectural separation of identity and transaction data, raise the question of whether the trilemma is permanent or merely a reflection of the current state of technology. This paper argues that it is partially both. Cryptographic advances can shift the privacy–compliance frontier outward: a zero-knowledge proof that verifies a transaction meets AML thresholds without revealing the sender’s identity would allow higher privacy at the same compliance level, effectively loosening the trilemma’s tightest edge. However, three considerations suggest the trilemma cannot be fully dissolved by technology alone.

First, the privacy–stability edge is not primarily a cryptographic problem. The risk that anonymous CBDC amplifies bank runs stems from the substitutability of CBDC and deposits, not from the traceability architecture; even a perfectly private CBDC would still compete with bank deposits during stress periods. Second, regulatory compliance is not a static target: as technology enables more privacy, regulators may raise compliance

requirements (e.g., extending AML rules to smaller transactions), maintaining the tension at a new equilibrium. Third, no privacy-preserving technology reviewed in the sample has been deployed at a national scale; the gap between cryptographic possibility and operational reality (throughput, latency, user comprehension) remains substantial. The trilemma should therefore be understood as technology-contingent but not technology-soluble: innovation can widen the zone of compromise but is unlikely to eliminate the underlying trade-off structure. In this sense, the trilemma is neither universal nor purely temporary; it is a constraint whose tightness varies with the state of technology, the stringency of regulation, and the risk tolerance of the issuing central bank. What the current literature captures is the trilemma in its 2019–2025 configuration; future reviews may find that specific edges have loosened or tightened as the field evolves.

Figure 2 illustrates this framework. The “zone of compromise” in the centre, characterized by tiered privacy, two-tier architecture, and zero interest, represents the design space where most central banks are currently positioning their CBDC proposals. The quantitative evidence (Table 8) aligns with this tendency: 71.4% of privacy-coded articles recommend tiered approaches, and the case study evidence (Table 10) shows that all four major CBDC deployments adopt two-tier architectures.

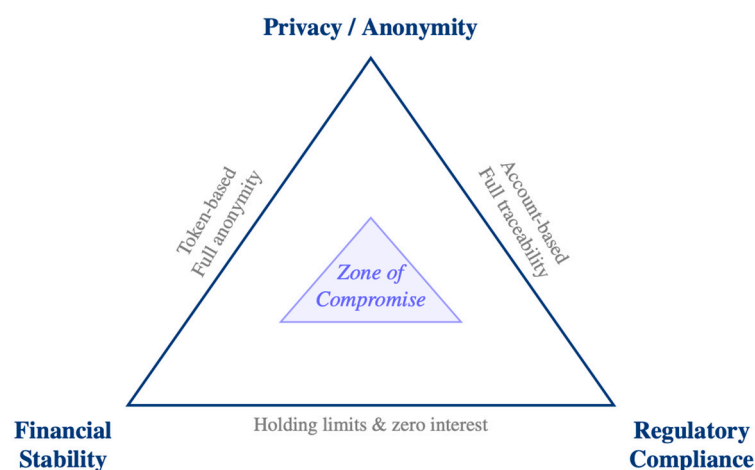


Figure 2. The CBDC design trilemma. Central banks must navigate trade-offs between privacy, financial stability, and regulatory compliance. The zone of compromise represents feasible designs (tiered privacy, two-tier architecture, zero interest) that partially satisfy all three objectives. No design identified in the 140-article sample simultaneously maximises all three vertices.

5.2. Cross-Theme Transmission Channels

The four themes are not independent research silos; they form a causal chain through which design choices propagate to outcomes. The evidence supports three principal transmission channels:

Design → Stability → Adoption (T1 → T2 → T4): Architectural choices (T1), particularly remuneration and holding limits, directly determine stability risk (T2). The safeguards adopted to manage that risk (zero interest, quantity caps) then reduce CBDC utility, weakening adoption appeal (T4). This channel explains why all four case study deployments show disappointing adoption: the stability safeguards that make CBDC “safe” simultaneously make it less attractive than existing payment alternatives.

Design → Privacy → Adoption (T1 → T3 → T4): Access model choices (T1) determine the degree of transaction traceability (T3), which in turn shapes user willingness to adopt (T4). Account-based designs enable compliance but deter privacy-sensitive users; token-based designs attract users but face regulatory resistance. Tiered privacy is the dominant compromise precisely because it attempts to interrupt this chain at a tolerable point.

Stability $\leftrightarrow$ Compliance feedback loop (T2 $\leftrightarrow$ T3): Stability safeguards (identity-linked holding limits) strengthen compliance capacity, but stringent compliance (extensive KYC, transaction monitoring) creates friction that can itself destabilize adoption and trigger the demand gap identified in advanced economies. This bidirectional channel produces the “regulatory paradox” of Theme 4: the strongest regulators move slowest because they internalize the compliance costs that weaker jurisdictions externalize.

5.3. Design Choice—Trade-Off Matrix

Table 11 maps eight key design dimensions against four outcome dimensions, synthesizing findings across all four themes.

Table 11. Design choice—trade-off matrix.

Design Dimension	Financial Stability	Privacy Outcome	Adoption Likelihood	Financial Inclusion	Key Sources
Account-based	Moderate risk	Low	Lower	High	Kahn et al. (2018); Auer and Böhme (2020)
Token-based	Higher risk	High	Variable	Lower	Kahn et al. (2005); Ok et al. (2025)
Hybrid	Manageable	Medium	Higher	Medium	Lamberty et al. (2024); Agur et al. (2022)
Interest-bearing	Risk	Neutral	Higher	High	Agur et al. (2022); Keister and Sanches (2022)
Non-interest	Lower risk	Neutral	Lower	Medium	Pfister (2024); Parra-Polania and Martinez-Ventura (2026)
Holding limits	Protective	Neutral	Lower	Lower	Fegatelli (2022); Parra-Polania and Martinez-Ventura (2026)
Two-tier arch.	Protective	Medium	Higher	Higher	Opore and Kim (2020); Grosman et al. (2024)
Offline capable	Neutral	Higher	Higher	Highest	Bibi and Yerzhan (2025); Lamberty et al. (2024)

Two patterns in Table 11 deserve emphasis. First, no row contains uniformly positive ratings across all four outcome columns; every design choice that improves one dimension degrades at least one other, confirming the trilemma at the level of individual design decisions. Second, the two designs rated “Protective” for stability (holding limits and two-tier architecture) both constrain privacy or adoption, while the two designs rated highest for privacy and adoption (token-based and offline-capable) introduce stability or compliance risks. This cross-column asymmetry operationalizes the trilemma: a policymaker can use Table 11 to trace how moving toward any vertex (e.g., choosing token-based for privacy) simultaneously moves away from the other two.

5.4. Answering the Research Questions

5.4.1. PRQ: How Do CBDC Design Choices Influence Trade-Offs?

CBDC design choices create a multi-dimensional trade-off space where no single design optimally serves all objectives. The literature converges on four key findings:

1. Two-tier hybrid architectures with tiered privacy represent the dominant recommendation in the sample, balancing competing objectives through graduated anonymity and intermediated distribution.

2. Financial stability concerns dominate design constraints; features like holding limits and zero remuneration are adopted primarily to protect banks, even though they reduce CBDC utility.
3. Privacy is the pivotal trade-off dimension, connecting adoption (privacy concerns deter users), stability (anonymity enables money laundering), and inclusion (simplified KYC enables unbanked access).
4. Context matters decisively; optimal design varies by institutional quality, financial system structure, digital literacy, and existing payment infrastructure.

5.4.2. SQ1: What Factors Drive or Hinder CBDC Adoption?

The five strongest adoption drivers are: (1) trust in issuing institution, (2) perceived usefulness, (3) financial literacy, (4) financial-inclusion motivation, and (5) social influence. The five strongest barriers are: (1) privacy and surveillance concerns, (2) perceived risk, (3) switching costs, (4) lack of awareness, and (5) digital divide. Advanced economies face switching-cost barriers, while developing economies face infrastructure and literacy barriers.

5.4.3. SQ2: How Do Privacy–Security Trade-Offs Affect Adoption?

Privacy concerns have a statistically significant negative effect on adoption intention across multiple survey studies. The relationship is mediated by trust: higher perceived privacy leads to higher trust and higher adoption willingness. Tiered privacy represents the compromise position, achieving moderate trust and acceptable adoption levels. Users apply a privacy calculus, weighing perceived benefits against perceived privacy costs.

Critically, however, privacy matters only up to the point where it conflicts with usability and fraud protection. While consumers express great concern for data protection, they consistently exhibit a preference for payment instruments that provide fund recoverability and budget monitoring. The demand for anonymity is often secondary to the demand for financial safety: users are willing to accept traceability if it provides mechanisms to dispute transactions, reverse fraudulent payments, and avoid the “irrevocable loss” associated with anonymous bearer instruments (Abramova et al., 2022; Choi et al., 2024). This “privacy paradox”, the mismatch between stated preferences and actual behaviour, is reinforced by “privacy numbness” from repeated data breaches, which desensitizes users to surveillance concerns (Feri et al., 2016). Experimental evidence confirms that privacy design variations have large effects on adoption for privacy-sensitive goods (up to 64% increase) but negligible effects for routine purchases (Choi et al., 2024), demonstrating that privacy is a conditional, rather than absolute, determinant of adoption.

5.4.4. SQ3: Which Architectures Best Support Inclusion Without Destabilizing Banking?

The optimal architecture for financial inclusion with stability preservation is a two-tier intermediated system with: (a) simplified KYC for small-value wallets, (b) zero remuneration, (c) offline capability, and (d) holding limits. Evidence from the Bahamas, Nigeria, and China suggests that while these designs are technically sound, adoption outcomes depend on complementary factors including merchant acceptance, digital literacy, and institutional trust.

5.5. Research Gaps

This review identifies nine significant research gaps (Table 12).

The gaps in Table 12 are not random omissions; they cluster around the weakest-evidenced parts of the trilemma. The three “High”-priority gaps (empirical evidence, cross-border interoperability, cybersecurity) all concern edges of the trilemma that are currently sustained by theoretical argument rather than observed data. If live-deployment studies were to show that disintermediation is modest (as China’s e-CNY pilot tentatively

suggests), the stability–privacy edge would loosen, and the trilemma’s practical binding force would diminish. Conversely, if cybersecurity vulnerabilities prove severe in offline token designs, the privacy–compliance edge tightens further. The trilemma is thus not a static framework but one whose shape depends on filling precisely these empirical gaps. A further nuance deserves emphasis: the few existing empirical studies rely predominantly on the Chinese e-CNY pilot, which operates within a specific institutional, legal, and geopolitical context. Generalizing findings from this single deployment to universal policy conclusions risks overstating the robustness of the evidence. Future empirical work should urgently collect and analyse data from Western democratic contexts and other legal systems to test whether the trilemma’s edges hold with the same intensity across different institutional environments.

Table 12. Identified research gaps.

Research Gap (Area)	Description of Gap and Evidential Basis	Priority Level
Empirical evidence	59% of articles are theoretical; few analyse live deployment data	High
Wholesale CBDC	Only 1 article focuses on wholesale CBDC	Medium
Cross-border interoperability	Limited analysis of multi-CBDC arrangements	High
Long-term effects	Most models analyse short-term introduction effects	Medium
Vulnerable populations	Limited research on elderly, rural, and low-income users	High
Cybersecurity	Only 1 article addresses CBDC-specific cyber risks	High
Legal frameworks	Few systematic cross-jurisdictional legal comparisons	Medium
Stablecoins/DeFi interaction	Limited analysis of CBDC competition with private digital currencies	Medium
Environmental impact	Only 1 article addresses energy implications	Low

5.6. Contradictions in the Literature

Five notable contradictions emerge from the synthesis:

1. Disintermediation severity: Some studies predict significant deposit outflows and credit crunches, while others argue that proper design safeguards make disintermediation risk minimal.
2. Privacy necessity: Some argue privacy is essential for adoption (users will reject traceable CBDC), while others contend that transparency benefits outweigh privacy costs.
3. Interest-bearing CBDC: Proponents argue interest strengthens monetary transmission and improves welfare; opponents warn it dangerously competes with deposits.
4. Financial inclusion: Some find that CBDC significantly improves inclusion; others argue that digital literacy requirements may worsen the digital divide.
5. Institutional quality: Strong institutions are expected to facilitate CBDC development, yet empirical evidence shows that strong regulators paradoxically move more slowly.

These contradictions highlight that CBDC outcomes are highly context-dependent and underscore the importance of jurisdiction-specific analysis.

6. Conclusions

6.1. Key Contributions

This systematic literature review of 140 peer-reviewed articles set out to examine a central proposition: that CBDC design is constrained by a structural trilemma among

privacy, financial stability, and regulatory compliance. The evidence is consistent with this proposition across all three edges.

First, this review proposes the CBDC design trilemma as a unifying analytical framework and finds the literature consistent with it. Cross-tabulation of coded variables shows that privacy-enhancing designs tend to weaken regulatory compliance, anonymous holdings are associated with amplified disintermediation risk, and stringent stability safeguards constrain transaction monitoring. No article in the sample describes a design that fully optimizes all three vertices, and the convergence on tiered-privacy, two-tier architectures reflects the kind of interior compromise the trilemma would predict.

Second, by synthesizing evidence across four thematic dimensions rather than within isolated streams, the review reveals cross-dimensional trade-offs invisible to single-theme analyses, most notably that privacy is the pivotal vertex connecting adoption demand (T4), stability risk (T2), and compliance feasibility (T3).

Third, the design choice trade-off matrix (Table 11) operationalizes the trilemma for policy use, enabling policymakers to trace how a specific design decision (e.g., raising holding limits) propagates across all three vertices.

6.2. Policy Implications

The findings carry several implications for CBDC policymakers:

CBDC design should be treated as a multi-objective optimization problem, not a technical implementation task. Design choices must be evaluated against all four dimensions simultaneously.

- Tiered privacy with simplified KYC for small transactions represents the most viable compromise between privacy, compliance, and inclusion.
- Stability safeguards (holding limits, zero interest) are essential but come at the cost of reduced CBDC utility; policymakers must communicate these trade-offs transparently.
- Financial literacy programmes should accompany CBDC deployment to maximize adoption and inclusion benefits.
- International coordination on cross-border CBDC interoperability deserves urgent attention, given the identified research gap.

6.3. Limitations

This review has several limitations. First, although the systematic search was conducted primarily through the Web of Science and supplemented by Scopus and SSRN, other databases (e.g., Google Scholar, IEEE Xplore) and grey-literature repositories may contain additional relevant studies not captured here. Second, the predominantly theoretical nature of the literature (59.3%) limits the strength of empirical conclusions. Third, the rapid pace of CBDC development means that new evidence from pilot programmes may quickly alter the landscape.

Fourth, and most consequential for the trilemma findings, the robustness of the coded variables to coder bias deserves explicit discussion. Two fields are particularly susceptible: “financial stability impact” (coded as positive, negative, mixed, or NA) and “relevance to PRQ” (coded as high, medium, or low). Both require interpretive judgement, and a different coder might assign different values, particularly for articles whose stability findings are conditional on design parameters (coded here as “mixed” but potentially classifiable as “positive” or “negative” depending on which design scenario is emphasized). Although intra-coder reliability was assessed (Cohen’s $\kappa = 0.81$), this measures consistency within a single coder, not accuracy against a ground truth. Independent dual coding would strengthen the robustness of the extraction. To assess sensitivity, readers should note that the core trilemma finding that no article proposes a design simultaneously maximizing all

three vertices does not depend on the stability or relevance codings; it rests on the *absence* of a particular design type in the sample, which is a more robust observation than any individual coding judgement. The quantitative cross-tabulations (Tables 6–9), however, are directly sensitive to coding choices and should be interpreted accordingly. Future systematic literature reviews in the CBDC field should consider independent dual coding by at least two researchers as a methodological standard, in line with best practice for SLR research synthesis.

Fifth, coding based on abstracts, introductions, and key findings sections may not capture the full nuance of some articles.

Sixth, the near absence of wholesale CBDC from this review is a significant scope limitation. Only 1 of the 140 articles focuses exclusively on wholesale CBDC. It is worth noting that a further 25 articles in the sample examine hybrid (retail and wholesale combined) CBDC variants; however, these articles treat the wholesale dimension only superficially, subordinating it to the retail analysis, and therefore do not provide sufficient insight to close this gap. Nevertheless, wholesale applications, cross-border settlement, interbank liquidity management, and multi-CBDC arrangements (e.g., mBridge) carry substantial implications for financial stability and geopolitical dynamics. The trilemma framework, as developed here, is calibrated primarily to retail CBDC, where the privacy–compliance tension concerns individual users, and the stability risk concerns deposit migration. Wholesale CBDC may involve a different trilemma entirely: one between settlement efficiency, monetary sovereignty, and cross-border interoperability. The wholesale gap is not a choice of this review but a reflection of the literature itself: the academic community has focused overwhelmingly on retail CBDC, leaving wholesale analysis to central bank working papers and policy reports that fall outside WOS-indexed journals. Future reviews should explicitly target wholesale and cross-border CBDC literature, potentially drawing on BIS, IMF, and central bank working paper series as primary sources.

6.4. Future Research Directions

Based on the identified gaps and contradictions, the following research directions are prioritized:

1. Empirical studies of live CBDC deployments: As more countries launch pilots, research should leverage transaction-level data to test theoretical predictions about disintermediation, adoption, and inclusion.
2. Cross-border CBDC frameworks: multi-CBDC arrangements (e.g., mBridge) require analysis of interoperability, currency competition, and geopolitical implications.
3. Cyber security risk assessment: Dedicated frameworks for evaluating CBDC-specific cyber threats are urgently needed. Specific technical areas requiring dedicated investigation include attacks on the network layer of distributed CBDC infrastructure; the resilience of offline hardware wallets to physical and side-channel exploits; and the long-term threat posed by quantum computing to the cryptographic primitives underpinning CBDC privacy and transaction integrity.
4. Vulnerable population impact: Longitudinal studies of CBDC effects on elderly, rural, and low-income users would strengthen the evidence base on inclusion.
5. Resolving contradictions: Comparative empirical studies across jurisdictions can help resolve contradictions about disintermediation severity, privacy necessity, and the institutional quality paradox.

Ultimately, the “cash paradox” that opened this review, rising demand for cash as a store of value alongside declining transactional use, is itself a manifestation of the design trilemma: cash offers near-perfect privacy but zero regulatory traceability and, at scale, potential stability risk. Whether central banks can construct a digital successor that

navigates this trilemma, preserving enough privacy to sustain public trust, enough stability safeguards to prevent disintermediation, and enough transparency to satisfy regulatory mandates will determine not only the success of individual CBDC projects but the future architecture of public money itself. The evidence reviewed here suggests that the trilemma cannot be dissolved, but it can be managed through tiered designs, jurisdictional calibration, and the ongoing empirical evaluation that the literature so urgently needs.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/ijfs14050122/s1>, Table S1: PRISMA 2020 Checklist and File S1: Full Search Strategy.

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References

- Abad, J., Nuño, G., & Thomas, C. (2025). CBDC and the operational framework of monetary policy. *Journal of Monetary Economics*, 151, 103762. [CrossRef]
- Abramova, S., Böhme, R., Elsinger, H., Stix, H., & Summer, M. (2022). *What can CBDC designers learn from asking potential users? Results from a survey of Austrian residents*. Working Paper No. 241. Oesterreichische Nationalbank. Available online: https://www.oenb.at/dam/jcr:e3199ed9-0b24-4df5-aac9-52c12c2fbc72/WP_241.pdf (accessed on 26 February 2026).
- Agur, I., Ari, A., & Dell’Ariccia, G. (2022). Designing central bank digital currencies. *Journal of Monetary Economics*, 125, 62–79. [CrossRef]
- Atangana, O., Khoukhi, L., Barbier, M., & Kokcam, A. (2026). Block PAD: A blockchain enabled framework for resilient and flexible CBDC transactions leveraging digital identity. *Computer Networks*, 275, 111805. [CrossRef]
- Atlantic Council. (2025, February). *CBDC tracker*. Available online: <https://www.atlanticcouncil.org/cbdctracker/> (accessed on 1 February 2025).
- Auer, R., & Böhme, R. (2020). The technology of retail central bank digital currency. *BIS Quarterly Review*. Available online: https://www.bis.org/publ/qtrpdf/r_qt2003j.htm (accessed on 26 February 2026).
- Auer, R., Frost, J., Gambacorta, L., Monnet, C., Rice, T., & Shin, H. S. (2022). Central bank digital currencies: Motives, economic implications, and the research frontier. *Annual Review of Economics*, 14(1), 697–721. [CrossRef]
- Bai, H., Cong, L. W., Luo, M., & Xie, P. (2025). Adoption of central bank digital currencies: Initial evidence from China. *Journal of Corporate Finance*, 91, 102735. [CrossRef]
- Barrdear, J., & Kumhof, M. (2021). The macroeconomics of central bank digital currencies. *Journal of Economic Dynamics and Control*, 142, 104148. [CrossRef]
- Beckmann, L., Debener, J., Hark, P. F., & Pfingsten, A. (2024). CBDC and the shadow of bank disintermediation: US stock market insights on threats and remedies. *Finance Research Letters*, 67, 105868. [CrossRef]

- Bibi, S., & Yerzhan, I. (2025). Central bank digital currency and digital payment instruments: Kazakhstan's experience between obstacles, threats and opportunities. *Journal of Post Keynesian Economics*, 48(3), 1–29. [CrossRef]
- Bijlsma, M., van der Cruysen, C., Jonker, N., & Reijerink, J. (2023). What triggers consumer adoption of central bank digital currency? *Journal of Financial Services Research*, 65(1), 1–40. [CrossRef]
- BIS. (2020). *Central bank digital currencies: Foundational principles and core features*. BIS Papers, No. 107. BIS. Available online: <https://www.bis.org/publ/othp33.htm> (accessed on 26 February 2026).
- Bitter, L. (2024). Banking crises under a central bank digital currency (CBDC). *Jahrbücher Für Nationalökonomie Und Statistik*, 245(4–5), 479–526. [CrossRef]
- Borgonovo, E., Caselli, S., Cillo, A., Masciandaro, D., & Rabitti, G. (2021). Money, privacy, anonymity: What do experiments tell us? *Journal of Financial Stability*, 56, 100934. [CrossRef]
- Broby, D. (2024). Central bank digital currencies—Lessons from China. *Journal of Information Science and Engineering*, 40(4), 699–715. [CrossRef]
- Chen, H., & Siklos, P. L. (2022). Central bank digital currency: A review and some macro-financial implications. *Journal of Financial Stability*, 60, 100985. [CrossRef]
- Choi, S., Kim, B., Kim, Y. S., & Kwon, O. (2024). Central bank digital currency and privacy: A randomized survey experiment. *International Economic Review*, 66(2), 823–847. [CrossRef]
- Chua, B.-L., Ayalew, A. Z., Kim, S., Han, H., & Ha, H. (2025). Consumer multifaceted trust of central bank digital currency (CBDC) payment in travel and tourism. *Technological Forecasting and Social Change*, 217, 124166. [CrossRef]
- Davoodalhosseini, S. M., & Rivadeneyra, F. (2020). A policy framework for E-money. *Canadian Public Policy*, 46(1), 94–106. [CrossRef]
- Demertzis, M., & Martins, C. (2023). *The value added of central bank digital currencies: A view from the euro area*. Policy Brief 13/2023. Bruegel. Available online: <https://www.bruegel.org/policy-brief/value-added-central-bank-digital-currencies-view-euro-area> (accessed on 26 February 2026).
- Desai, R. (2024). Analysing consumer switching behaviour and financial antecedents to determine the intention to use central bank digital currency: Evidence from the Indian economy. *Asia Pacific Journal of Marketing and Logistics*, 37(7), 2048–2074. [CrossRef]
- Didenko, A. N., & Buckley, R. P. (2022). Central bank digital currencies as a potential response to some particularly Pacific problems. *Asia Pacific Law Review*, 30(1), 44–69. [CrossRef]
- Dionysopoulos, L., Marra, M., & Urquhart, A. (2023). Central bank digital currencies: A critical review. *International Review of Financial Analysis*, 91(91), 103031. [CrossRef]
- Elsayed, A. H., & Nasir, M. A. (2022). Central bank digital currencies: An agenda for future research. *Research in International Business and Finance*, 62, 101736. [CrossRef]
- Fegatelli, P. (2022). A central bank digital currency in a heterogeneous monetary union: Managing the effects on the bank lending channel. *Journal of Macroeconomics*, 71, 103392. [CrossRef]
- Feri, F., Giannetti, C., & Jentzsch, N. (2016). *Disclosure of personal information under risk of privacy shocks* (2nd ed.). Working paper No. 1055. Dipartimento di Scienze Economiche, Università di Bologna. [CrossRef]
- Fernández-Villaverde, J., Sanches, D., Schilling, L., & Uhlig, H. (2020). Central bank digital currency: Central banking for all? *Review of Economic Dynamics*, 41(1), 225–242. [CrossRef]
- Ferrari Minesso, M., Mehl, A., & Stracca, L. (2022). Central bank digital currency in an open economy. *Journal of Monetary Economics*, 127, 54–68. [CrossRef]
- Fujiki, H. (2023). Central bank digital currency, crypto assets, and cash demand: Evidence from Japan. *Applied Economics*, 56(19), 1–19. [CrossRef]
- Garratt, R., & van Oordt, M. R. C. (2021). Privacy as a public good: A case for electronic cash. *Journal of Political Economy*, 129(7), 2157–2180. [CrossRef]
- Geva, B., Grünewald, S. N., & Zellweger-Gutknecht, C. (2021). The e-Banknote as a “Banknote”: A monetary law interpreted. *Oxford Journal of Legal Studies*, 41(4), 1119–1148. [CrossRef]
- Grosman, A., Sayyid, A., & Klarin, A. (2024). Central bank digital currencies and institutional systems: Design choices based on institutional characteristics. *European Journal of International Management*. [CrossRef]
- Heitmann, D., Koch, J.-A., Islam, M. S., & Eva, S. A. (2025). The impact of central bank digital currencies on the financial stability of banks: Dynamic panel estimation. *Finance Research Letters*, 84, 107791. [CrossRef]
- Hoang, Y. H., Ngo, V. M., & Bich Vu, N. (2023). Central bank digital currency: A systematic literature review using text mining approach. *Research in International Business and Finance*, 64, 101889. [CrossRef]
- Hua, Q., Qiu, Z., Jiang, T., & Tang, K. (2025). Macroeconomic effects of CBDC negative interest policy in an open economy: A comparison of quantity and price rules. *International Review of Economics & Finance*, 100, 104119. [CrossRef]
- Islam, M. M., & IN, H. P. (2022). A privacy-preserving transparent central bank digital currency system based on consortium blockchain and unspent transaction outputs. *IEEE Transactions on Services Computing*, 16, 2372–2386. [CrossRef]

- Jabbar, A., Geebren, A., Hussain, Z., Dani, S., & Ul-Durar, S. (2023). Investigating individual privacy within CBDC: A privacy calculus perspective. *Research in International Business and Finance*, 64, 101826. [\[CrossRef\]](#)
- Jiang, J. H., & Shao, E. (2019). The cash paradox. *Review of Economic Dynamics*, 36, 177–197. [\[CrossRef\]](#)
- Jin, S. Y., & Xia, Y. (2022). CEV framework: A central bank digital currency evaluation and verification framework with a focus on consensus algorithms and operating architectures. *IEEE Access*, 10, 63698–63714. [\[CrossRef\]](#)
- Kahn, C. M., McAndrews, J., & Roberds, W. (2005). Money is privacy. *International Economic Review*, 46(2), 377–399. [\[CrossRef\]](#)
- Kahn, C. M., Rivadeneyra, F., & Wong, T.-N. (2018). *Should the central bank issue E-money?* Bank of Canada staff working paper, 2018-58. Bank of Canada. [\[CrossRef\]](#)
- Kang, K.-Y. (2024). Digital currency and privacy. *Theoretical Economics*, 19(1), 131–167. [\[CrossRef\]](#)
- Keister, T., & Sanches, D. (2022). Should central banks issue digital currency? *The Review of Economic Studies*, 90(1), 404–431. [\[CrossRef\]](#)
- Kim, J. J., Kim, S., Hailu, T. B., Ryu, H. B., & Han, H. (2023). Does central bank digital currency (CBDC) payment create the opportunity for the tourism industry? *Journal of Hospitality & Tourism Research*, 48(6), 109634802311587. [\[CrossRef\]](#)
- Kim, Y. S., & Kwon, O. (2022). Central bank digital currency, credit supply, and financial stability. *Journal of Money, Credit and Banking*, 55(1), 297–321. [\[CrossRef\]](#)
- Kuehnlenz, S., Orsi, B., & Kaltenbrunner, A. (2022). Central bank digital currencies and the international payment system: The demise of the US dollar? *Research in International Business and Finance*, 64, 101834. [\[CrossRef\]](#)
- Lamberty, R., Kirste, D., Kannengießner, N., & Sunyaev, A. (2024). HybCBDC: A design for central bank digital currency systems enabling digital cash. *IEEE Access*, 12, 137712–137728. [\[CrossRef\]](#)
- Li, J. (2022). Predicting the Demand for central bank digital currency: A structural analysis with survey data. *Journal of Monetary Economics*, 134, 73–85. [\[CrossRef\]](#)
- Liu, X., Wu, J., & Zhang, C. (2025). Antecedents of consumers' acceptance of central bank digital currency: The role of technology perceptions, social influence and personal traits. *Technological Forecasting and Social Change*, 217, 124192. [\[CrossRef\]](#)
- Ma, C., Jin, Z., Mei, Z., Zhou, F., She, X., Huang, J., & Liu, D. (2022). Internet of things background: An empirical study on the payment intention of central bank digital currency design. *Mobile Information Systems*, 2022, 4846372. [\[CrossRef\]](#)
- Mertzanis, C., Ahmed, M. S., Faisal, R., & Al Suwaidi, R. A. (2025). Does financial literacy drive digital currency innovation? cross-country evidence on CBDC adoption. *Borsa Istanbul Review*, 25, 109–118. [\[CrossRef\]](#)
- Michalopoulos, P., Olowookere, O., Pocher, N., Sedlmeir, J., Veneris, A., & Puri, P. (2025). Privacy and compliance design options in offline central bank digital currencies. *IEEE Transactions on Network and Service Management*, 22(5), 3748–3763. [\[CrossRef\]](#)
- Milne, A. (2023). Argument by false analogy: The mistaken classification of Bitcoin as token money. *Journal of Money, Credit and Banking*, 56(8), 2199–2222. [\[CrossRef\]](#)
- Moro, A., & Nispi Landi, V. (2024). The external financial spillovers of CBDCs. *Journal of Economic Dynamics & Control*, 159, 104801. [\[CrossRef\]](#)
- Niepelt, D. (2024). Money and banking with reserves and CBDC. *The Journal of Finance*, 79(4), 2505–2552. [\[CrossRef\]](#)
- Ok, T.-M., Shin, J.-S., Kim, J.-S., & Choi, D.-S. (2025). Applying UTXO-based privacy-preserving technologies to an account-based CBDC Architecture. *Journal of Internet Technology*, 26(5), 557–577. [\[CrossRef\]](#)
- Opore, E. A., & Kim, K. (2020). A compendium of practices for central bank digital currencies for multinational financial infrastructures. *IEEE Access*, 8, 110810–110847. [\[CrossRef\]](#)
- Osakwe, C. N., Ogunmokun, O. A., Elgammal, I., & Kwarteng, M. A. (2024). Individuals' attitudes and their adoption intentions of central bank digital currency: Combining theories and analytics for deeper insights. *Technological Forecasting and Social Change*, 210, 123857. [\[CrossRef\]](#)
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *Journal of Clinical Epidemiology*, 134, 178–189. [\[CrossRef\]](#)
- Park, J. (2024). Large excess reserves, central bank digital currency, and monetary policy. *Macroeconomic Dynamics*, 29, 1–23. [\[CrossRef\]](#)
- Parra-Polania, J. A., & Martinez-Ventura, C. (2026). Optimal CBDC design: A model with two access mechanisms and the role of anonymity. *Journal of Financial Stability*, 83, 101516. [\[CrossRef\]](#)
- Pfister, C. (2024). For a political economy of central bank digital currency. *Revue d'Économie Politique*, 134(4), 525–551. [\[CrossRef\]](#)
- Pocher, N., & Veneris, A. (2021). Privacy and transparency in CBDCs: A regulation-by-design AML/CFT scheme. *IEEE Transactions on Network and Service Management*, 19(2), 1776–1788. [\[CrossRef\]](#)
- Radic, A., Quan, W., Koo, B., Chua, B.-L., Kim, J. J., & Han, H. (2022). Central bank digital currency as a payment method for tourists: Application of the theory of planned behavior to digital Yuan/Won/Dollar choice. *Journal of Travel & Tourism Marketing*, 39(2), 152–172. [\[CrossRef\]](#)
- Sanchez-Roger, M., & Puyol-Antón, E. (2021). Digital bank runs: A deep neural network approach. *Sustainability*, 13(3), 1513. [\[CrossRef\]](#)

- Sandhu, K., Dayanandan, A., & Kuntluru, S. (2023). India's CBDC for digital public infrastructure. *Economics Letters*, 231, 111302. [\[CrossRef\]](#)
- Schilling, L., Fernández-Villaverde, J., & Uhlig, H. (2024). Central bank digital currency: When price and bank stability collide. *Journal of Monetary Economics*, 145, 103554. [\[CrossRef\]](#)
- Sethaput, V., & Innet, S. (2023). Blockchain application for central bank digital currencies (CBDC). *Cluster Computing*, 26, 2183–2197. [\[CrossRef\]](#)
- Sha, C., Che, T., Xu, T., & Yang, Z. (2024). Antecedents of users' switching intention to Central Bank Digital Currency: A push-pull-mooring model perspective. *Electronic Commerce Research and Applications*, 68, 101467. [\[CrossRef\]](#)
- Son, J., Bilgin, M. H., & Ryu, D. (2022). Consumer choices under new payment methods. *Financial Innovation*, 8(1), 82. [\[CrossRef\]](#)
- Stroukal, D., & Peterka, P. (2025). Tokenization as a pathway to anonymity in central bank digital currencies. *Research in International Business and Finance*, 75, 102769. [\[CrossRef\]](#)
- Takaragi, K., Kubota, T., H., S., Umezawa, K., & Koyanagi, H. (2022). Secure revocation features in eKYC—Privacy protection in central bank digital currency. *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, 106, 325–332. [\[CrossRef\]](#)
- Tronnier, F., Harborth, D., & Hamm, P. (2022). Investigating privacy concerns and trust in the digital Euro in Germany. *Electronic Commerce Research and Applications*, 53, 101158. [\[CrossRef\]](#)
- Tsang, C.-Y., Yang, Y.-P., & Chen, P.-K. (2023). Disciplining CBDCs: Achieving the balance between privacy protection and central bank independence. *Northwestern Journal of International Law & Business*, 43(3), 235.
- Tunç, H. (2026). Why strong regulators move slowly: A global paradox in central bank digital currency adoption. *Economics Letters*, 262, 112863. [\[CrossRef\]](#)
- Tunzina, T., Chayon, A. K., Jitu, P. G., Ankon, M. U., Joy, S. N., Shaha, R. K., Islam, M., Hussain, M. S., Hassan, M. M., & Pham, P. H. (2024). Blockchain-based central bank digital currency: Empowering centralized oversight with decentralized transactions. *IEEE Access*, 12, 192689–192709. [\[CrossRef\]](#)
- Umar, M. (2025). The role of financial literacy in accelerating the adoption of a central bank digital currency. *Borsa Istanbul Review*, 25, 101517. [\[CrossRef\]](#)
- Westermeier, C. (2022). From flows towards updates: Security regimes and changing technologies for financial surveillance. *Review of International Studies*, 49(4), 615–636. [\[CrossRef\]](#)
- Wu, J., Liu, X., & Zhang, C. (2024). Unveiling the influencing mechanism underlying users' adoption and recommend intentions of central bank digital currency: A behavioral reasoning theory perspective. *Journal of Retailing and Consumer Services*, 81, 104050. [\[CrossRef\]](#)
- Xiao, Z., Niu, Y., Xiang, L., & Cui, Y. (2024). Central bank digital currency and corporate debt financing costs: Empirical evidence from China. *Finance Research Letters*, 69, 106195. [\[CrossRef\]](#)
- Yang, Q., Zhou, G., & Chao, X. (2025). Forecasting central bank digital currency demand from mobile payment behaviour: Evidence from Chinese family financial literacy survey. *Spanish Journal of Finance and Accounting/Revista Española de Financiación Y Contabilidad*, 54(3), 319–346. [\[CrossRef\]](#)
- Yue, W. (2025). Subduing Leviathan: Examining Central Bank Digital Currency's privacy protection regime from the Chinese and European Perspectives. *Asia Pacific Law Review*, 33(2), 207–230. [\[CrossRef\]](#)
- Zhang, G., & Chan, K. C. (2025). Central bank digital currency and corporate cash holdings: Evidence from China's e-CNY pilot. *Journal of Corporate Finance*, 94, 102847. [\[CrossRef\]](#)
- Zhang, J., Tian, R., Cao, Y., Yuan, X., Yu, Z., Yan, X., & Zhang, X. (2021). A hybrid model for central bank digital currency based on blockchain. *IEEE Access*, 9, 53589–53601. [\[CrossRef\]](#)
- Zhang, T., & Huang, Z. (2021). Blockchain and central bank digital currency. *ICT Express*, 8(2), 264–270. [\[CrossRef\]](#)

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